

Master-Slave Synchronous Control of Dual-Drive Gantry Stage With Cogging Force Compensation

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Abstract—Dual-drive gantry stage has been widely applied to various industrial manufacturing fields with its unique structural advantages, and the synchronous control accuracy of the platform is crucial to the performance of the whole motion system. Therefore, an adaptive robust synchronous control scheme based on an improved master-slave structure is proposed, which is not only simple in structure but also easy to implement in engineering. The error dynamics model established in this article makes up for the lag of response of traditional master-slave control and improves the stability of closed-loop system. Online parameter adaptive algorithms deal with parameter uncertainties in the system, while robust control deals with unmodeled dynamics and external disturbances. In addition, nonlinear cogging force compensation is applied to the gantry biaxial system to further improve the control accuracy of tracking and synchronization. Finally, a dual-drive gantry stage system with good tracking and synchronization performance is obtained. The effectiveness and superiority of the proposed control strategy are verified by the comparison of several groups of experiments.

Index Terms—Adaptive robust control (ARC), cogging force compensation, master-slave control, synchronization control.

I. INTRODUCTION

IN MODERN industry, the H-type gantry stage as a typical mechanical structure has the advantages of strong mechanical coupling, large thrust, and high rigidity. It has been widely used in high-end manufacturing equipment with typical performance requirements, such as chip mounter, lithography machine, laser cutting machine, optical detection system, etc. A dual-drive gantry stage can provide high speed, high precision, and high dynamic performance, which has gradually become the mainstream structure in many fields. Many

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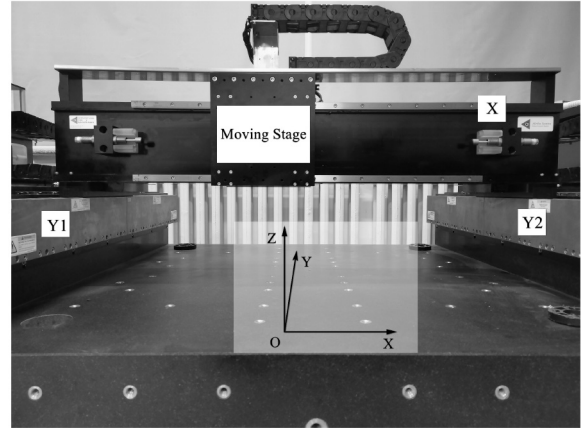


Fig. 1. Dual drive gantry stage.

significant studies have been carried out on such a system in recent years [1]–[7]. The structure of the gantry stage is displayed in Fig. 1. Two linear motor modules are, respectively, installed at both ends of the gantry stage, and two sliders on the linear guide rail are rigidly connected with the beam. The cross beam with two linear motors jointly driven moves along the direction of the linear guide rail, while the moving stage on the beam moves along the X direction. Generally speaking, all kinds of production tasks are accomplished by high-precision positioning control or high-precision tracking control of each axis.

As is known to all, the realization of accurate synchronization and stable tracking is the most important problem in the control system of the dual drive gantry stage. However, the flexible deformation of the ball bearing in the linear guide will cause the vibration of the beam and lead to the instability of the gantry stage system. In addition, due to the influence of load variation, parameter uncertainties, and unknown disturbances, it brings great challenges to realize high precision synchronous control of the system. Once the two y -axis servo units are out of synchronization in the process of movement, the beam will rotate around its center of gravity, which will directly destroy the stability of the system. It is nevertheless noted that the strong mechanical coupling and mutual restriction between the beam and the two linear guides will produce greater internal force and resistance on the beam. When moving at high speed and high acceleration, the beam easily exceeds the physical limit of the gantry stage, which can not guarantee the normal operation of the platform, and even cause direct damage to the machinery. It can be seen that

the high precision synchronization of the gantry table is the prerequisite for the system to obtain excellent performance.

Many representative synchronous control strategies have been proposed to solve this challenge in recent years [8], which can be summarized as a parallel control strategy, series control strategy, cross-coupling control strategy, and direct rotation control strategy. In the parallel synchronization strategy, the control command is sent to each servo unit at the same time, and each axis tracks the reference trajectory under the condition that the parameters match as much as possible. Although the parallel control strategy has a simple structure, its anti-interference capability is poor because there is no connection between servo control units. A dual-drive gantry stage model based on the decoupling scheme was established to reduce the influence of mechanical coupling on synchronization accuracy and improve system performance in [9]. In the serial control strategy, the master servo controller starts driving output after receiving the reference instruction, and transmits the output information to the slave servo controller for driving control. This kind of control strategy usually makes the dynamic response of the slave servo unit lag, which leads to poor synchronization performance in the process of system motion. A master-slave synchronous control method for CNC machine tools is proposed in [10]. The error dynamics equation is obtained through the reference model, and the variable structure controller is designed to reduce the synchronization error. The cross-coupling control strategy is a kind of control which couplings the tracking error and synchronization error in order to realize the synchronization of two axes. This control strategy provides good anti-interference ability, but it is difficult to select appropriate coupling coefficient and extend it to the multiaxis system. In [11], a synchronous control scheme combining the second-order discrete time-fractional sliding mode algorithm with cross-coupling control is implemented to reduce chattering and improve the control accuracy of the system. A model predictive synchronous control scheme based on cross-coupling strategy and an optimal PID parameter design synchronous control scheme were proposed in [12] and [13], both of which have achieved certain effects. Direct consideration of the rotation dynamics control is a strategy to transform the synchronization control of two drive shafts into the rotation control of crossbeam to obtain good synchronization performance. In [14] and [15], a good synchronization control effect is achieved by analyzing the internal cause of the nonsynchronization of two drive shafts and directly controlling the rotation of the beam. However, the design process of the control algorithm is too complicated, which is not conducive to the implementation of the project. Therefore, it is of great engineering significance to ensure the simple structure and easy implementation of the dual-drive gantry synchronous control scheme.

In addition, it is necessary to choose a suitable control algorithm in an electromechanical control system. Neural network control, robust control, adaptive control, iterative learning, and feedback control [16]–[23] have been studied and applied in many industrial applications. In [24], a reactive control strategy is presented for the hybrid floating power generation platform, which significantly improves the performance of the

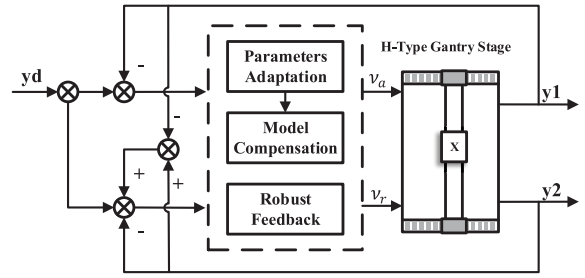


Fig. 2. Improved master-slave control structure for the dual-drive gantry stage.

hybrid energy system. For the uncertainties of system parameters, Yao and Jiang [25] proposed an adaptive robust control (ARC) framework combining the advantages of robust control and adaptive control. This method is very suitable for a class of nonlinear systems with variable parameters and uncertain models, so that the stability and anti-interference ability of the system can be guaranteed. Furthermore, nonlinear cogging force as an important interference factor needs to be fully considered in order to obtain higher control precision of the motor system. Many scholars have done a lot of meaningful work on the modeling and compensation of cogging forces in [26] and [27]. It should be pointed out that these studies on the compensation of cogging force rarely involve two-axis synchronous control systems.

This article mainly studies the precision of the dual-drive gantry stage and presents an improved master-slave synchronous control scheme with cogging force compensation. The main contributions are listed as follows.

- 1) Considering the complexity of the synchronous controller and the performance of the system, a simple improved master-slave structure is proposed by establishing an error dynamic model.
- 2) Due to the existence of unknown disturbances, load variations, and model uncertainties, an adaptive robust controller is designed to minimize the influence of these nonlinear factors and improve the system stability.
- 3) The compensation of nonlinear cogging force is fully considered and applied to the biaxial system to further improve the synchronization accuracy.

The remainder is designed as follows. In Section II, the structure of the master-slave control is analyzed in detail and the dynamic model of the gantry stage is established. The design steps of ARC and the stability analysis of the system are introduced in Section III. In Section IV, comparative experiments are carried out. Finally, a brief summary of this article in Section V.

II. PROBLEM FORMULATION

A. Improved Master-Slave Structure

In this section, an improved parallel master-slave control framework is proposed as shown in Fig. 2. The framework adopts the form of parallel command input, which avoids the lag effect of linear motor, and establishes feedback between master servo unit and slave servo unit, which enhances the disturbance rejection capability of the system. The position instructions received by the controller are simultaneously

applied to the two drive units, and the beam moves under the action of the drive shaft. When the synchronization error occurs due to external interference, the adaptive robust controller will adjust immediately to keep the synchronization between the two drive units as far as possible.

Let $y_d(t)$ stand for the input signal of the desired trajectory, while $y_1(t)$ and $y_2(t)$ represent the actual position information of the two axes returned by the encoder at any time in Fig. 2. Then, the tracking errors are written as $e_1 = y_1 - y_d$ and $e_2 = y_2 - y_d + (y_2 - y_1)$, respectively. Subsequently, the vector of synthetic error σ can be obtained by simplification, which is expressed as

$$\sigma = Pe, \quad P = \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix} \quad (1)$$

where $\sigma = [\sigma_1, \sigma_2]^T$ and $e = [e_1, e_2]^T$.

Remark 1: The lumped error σ reflects the comprehensive performance of the gantry motion system. It means that $y_1(t)$ needs to achieve zero error tracking of desired trajectory $y_d(t)$ as far as possible in finite time, while y_2 not only needs to track the desired trajectory perfectly, but also needs to consider the influence of tracking error of the master servo unit. That is, the $Y1$ axis guarantees excellent tracking performance under the control of the driver, while the $Y2$ axis achieves a better tradeoff between tracking performance and synchronization performance.

B. Dynamic Models

The effect of electrodynamics is expressed as disturbance by lumped parameter error, and the dynamic model of the whole gantry system in the Y direction can be simplified as

$$M\ddot{y} = F_m - F_f - F_c + d \quad (2)$$

where y denotes the position information of two linear motors, $\dot{y}$ and $\ddot{y}$, respectively, represent the current velocity and acceleration information. M represents the inertia coefficient of the whole beam and d is the lumped representation of various uncertain errors and unknown disturbances. F_m is the electromagnetic driving force and it can be expressed as $F_m = Kv$, where the former is the thrust coefficient related to the motor characteristics and the latter is the input voltage. F_f represents the friction generated on the linear guide rail during the movement of the system. Compared with many proposed friction models, a simple and generally applicable treatment is displayed as [28], which is a nonlinear function of velocity

$$F_f(\dot{y}) = B\dot{y} + A_f S_f(\dot{y}) \quad (3)$$

where B and A_f represent the coefficient of viscous damping and Coulomb friction at the guide rail, respectively. $S_f(\dot{y})$ can be expressed as $S_f(\dot{y}) = \arctan(\beta\dot{y})$ to replace the sign function $\text{sign}(\dot{y})$. Generally speaking, the value of β is designed to be large to approximate the characteristics of the sign function. F_c represents the cogging force in (2), which is usually assumed to be a periodic function of position in [26]. As a result, it can be introduced as follows:

$$F_c(y) = \sum_{i=1}^m \left(S_i \sin\left(\frac{2i\pi}{N}y\right) + C_i \cos\left(\frac{2i\pi}{N}y\right) \right) \quad (4)$$

where N is the distance between magnetic poles, S_i and C_i are constant coefficients. m represents the number of periodic functions required to approximate the cogging force. In this article, it is assumed that the magnetic pole distance N of the linear motor is equal. Therefore, the periodic function $F_c(y)$ with respect to the position y can be obtained by the above equation. It can be approximated by the sum of the following functions:

$$F_c(y) = A_c^T S_c(y) \quad (5)$$

where $A_c = [S_{11}, C_{11}, \dots, S_{m1}, C_{m1}]^T \in R^{2m}$ and $S_c(y) = [\sin([2\pi/N]y), \cos([2\pi/N]y), \dots, \sin([2m\pi/N]y), \cos([2m\pi/N]y)]^T$ represent the parameter vector to be designed and the function vector related to position, respectively. Therefore, considering the motion system of the dual-drive linear motor, the dynamics model can be established as

$$M_k \ddot{q} + B_k \dot{q} + A_{fk} S_f(\dot{q}) + A_{ck}^T S_c(q) = v + d_{nk} + \tilde{d} \quad (6)$$

where the serial numbers of the two y -axes are denoted by 1 and 2, respectively. Therefore, $q = [y_1(t), y_2(t)]^T$, $\dot{q} = [\dot{y}_1(t), \dot{y}_2(t)]^T$ and $\ddot{q} = [\ddot{y}_1(t), \ddot{y}_2(t)]^T$ are both 2×1 vector forms, which are expressed as the position, velocity, and acceleration information of the cross beam, respectively. In this article, it is considered that the electromagnetic thrust coefficients of the two linear motors are constant and $k_1 = k_2 = k$. For all items, there is $\bullet_k = \bullet/k$. $M_k = \text{diag}[M_{1k}, M_{2k}]$ and $B_k = \text{diag}[B_{1k}, B_{2k}]$ are 2×2 diagonal matrices representing the inertia vector and viscous damping coefficients, respectively; $A_{fk} = \text{diag}[A_{f1k}, A_{f2k}]$ is the coulomb friction coefficient, and $S_f(\dot{q}) = [S_f(\dot{y}_1), S_f(\dot{y}_2)]^T$; $A_{ck}^T = \text{diag}[A_{c1k}^T, A_{c2k}^T]$ and $S_c(q) = [S_c(y_1), S_c(y_2)]$; $v = [v_1, v_2]^T$ is the vector of actual control input; $d_{nk} = [d_{n1k}, d_{n2k}]^T$ represents the determined part in d and $\tilde{d} = d - d_{nk}$. In view of (1), the error dynamics equation can be inferred as

$$M_p \ddot{\sigma} + B_p \dot{\sigma} + M_y(\ddot{q}_d) + B_y(\dot{q}_d) + A_{fy} S_f(\dot{q}) + A_{cy}^T S_c(q) = v_p + d_p + \tilde{\Delta} \quad (7)$$

where $q_d = [y_d, y_d]$, $M_p = PM_k P^{-1}$, $B_p = PB_k P^{-1}$, $M_y = PM_k$, $B_y = PB_k$, $A_{fy} = PA_{fk}$, $A_{cy} = PA_{ck}$, $v_p = Pv$, $d_p = Pd_{nk}$, and $\tilde{\Delta} = P\tilde{d}$. In any finite workspace $q \in \Omega_q$, M_p is a positive definite symmetric matrix, we have

$$\delta_1 I \leq M_p \leq \delta_2 I \quad (8)$$

where δ_1 and δ_2 are positive diagonal matrices about the bound of M_p . The dynamics equation (7) can be expressed as a linearized vector of unknown parameters. Define the unknown parameter set as $\theta_r = [\theta_{r1}, \theta_{r2}, \theta_{r3}, \theta_{r4}, \theta_{r5}, \dots, \theta_{r(8+4m)}]^T = [M_{1k}, M_{2k}, B_{1k}, B_{2k}, A_{f1k}, A_{f2k}, A_{c1k}^T, A_{c2k}^T, d_{n1k}, d_{n2k}]^T$. Generally speaking, the range of parameter uncertainties can be obtained through system identification in practical application.

Assumption 1: The nominal values of system parameters and external disturbances are bounded, which can be described as follows:

$$\begin{aligned} \theta_r &= \Omega_{\theta_r} \triangleq \{\theta_r : \theta_{r\min} \leq \theta_r \leq \theta_{r\max}\} \\ \tilde{\Delta} &= \Omega_{\tilde{\Delta}} \triangleq \{\tilde{\Delta} : \|\tilde{\Delta}\| \leq \Delta_r\} \end{aligned} \quad (9)$$

where $\theta_{r \min} = [\theta_{r1 \min}, \theta_{r2 \min}, \theta_{r3 \min}, \dots, \theta_{r(8+4m) \min}]^T$, $\theta_{r \max} = [\theta_{r1 \max}, \theta_{r2 \max}, \theta_{r3 \max}, \dots, \theta_{r(8+4m) \max}]^T$, and Δ_r can obviously be approximated in practice.

III. ADAPTIVE ROBUST CONTROLLER DESIGN

In this section, referring to the sliding mode control method, an adaptive robust synchronization controller will be designed to make $q = [y_1(t), y_2(t)]^T$ track $q_d = [y_d(t), y_d(t)]$ as accurately as possible under the action of the control input v , and achieve excellent synchronization performance.

A. Adaptive Robust Control Law Synthesis

Based on the dynamics equation (7), a sliding surface function of tracking error is defined as

$$s = \dot{\sigma} + \kappa \sigma \quad (10)$$

where $\kappa = \text{diag}[\kappa_1, \kappa_2]$, κ_1 and κ_2 are positive constants. Therefore, it is easy to infer that the convergence of the tracking error σ is consistent with the sliding surface function s . Subsequently, a positive definite energy function can be designed as

$$V(t) = \frac{1}{2} s^T M_p s. \quad (11)$$

Differentiating (11) and substituting (7), we have

$$\dot{V}(t) = s^T [v_p + \psi_r(q, \dot{q}, t)\theta_r + \tilde{\Delta}] \quad (12)$$

where $\psi_r(q, \dot{q}, t)\theta_r = -M_y \ddot{q}_d - B_y \dot{q}_d - A_{fy} S_y(\dot{q}) - A_{cy}^T S_c(q) - B_p \dot{\sigma} + M_p \kappa \dot{\sigma} + d_p$, and $\psi_r(q, \dot{q}, t) = [\psi_{1r}, \psi_{2r}]$ is a $2 \times (8+4m)$ matrix, which can be obtained by analyzing the form of θ_r .

Considering the structure of (12), the ARC law is designed as follows:

$$v_p = v_a + v_r, \quad v_r = v_{r1} + v_{r2}. \quad (13)$$

Remark 2: For the adaptive control method, the designed controller mainly consists of three parts, model compensation term, error feedback term and nonlinear robust feedback term. $v_a = -\psi_r(q, \dot{q}, t)\hat{\theta}_r$ is the compensation term of the adjustable adaptive model, which eliminates the influence of system parameter uncertainties and realizes zero error tracking through online parameter self-adaptation. $\hat{\theta}_r$ and $\tilde{\theta}_r$ are the estimates and errors of θ_r , respectively (i.e., $\tilde{\theta}_r = \hat{\theta}_r - \theta_r$). $v_{r1} = -k_{r1}s$ is a simple proportional feedback, which is used to stable the nominal system. k_{r1} is a positive definite diagonal matrix that needs to be designed. v_{r2} is the nonlinear robust control term, which is applied to deal with various uncertainties.

Substituting (13) into (12) and simplifying the result, we have

$$\dot{V}(t) = s^T \left\{ v_r - [\psi_r(q, \dot{q}, t)\tilde{\theta}_r - \tilde{\Delta}] \right\}. \quad (14)$$

According to Assumption 1, the uncertainties of the system model in (14) can be limited by the following known functions:

$$|\psi_r(q, \dot{q}, t)\tilde{\theta}_r - \tilde{\Delta}| \leq \rho(q, \dot{q}, t) \quad (15)$$

where $\rho(q, \dot{q}, t) \geq \|\theta_{\max} - \theta_{\min}\| \|\psi_r\| + \Delta_r$. Therefore, the nonlinear robust term can generally be defined as $v_{r2} =$

$-\rho(q, \dot{q}, t)\text{sgn}(s)$ via the sliding mode control method, which guarantees the stability of the system, i.e., $\dot{V}(t) \leq -k_{r1}s^2 - \rho(q, \dot{q}, t)|s| + |s| |\psi_r(q, \dot{q}, t)\tilde{\theta}_r - \tilde{\Delta}| \leq 0$.

However, chattering linearity is inevitable due to the existence of discontinuous term $\text{sgn}(s)$. Define a smooth continuous function $Z(q, \dot{q}, t)$ to replace $\rho(q, \dot{q}, t)\text{sgn}(s)$, and satisfy the following conditions:

$$\begin{cases} i & -Z(q, \dot{q}, t)s \leq 0 \\ ii & \rho(q, \dot{q}, t)|s| - Z(q, \dot{q}, t)s \leq \varpi \end{cases} \quad (16)$$

where ϖ is an arbitrarily small constant representing the degree of approximation. An example of a smooth function is shown as

$$Z(q, \dot{q}, t) = \frac{1}{4\varpi} \rho^2(q, \dot{q}, t)s. \quad (17)$$

Obviously, $Z(q, \dot{q}, t)$ satisfies the constraint in (16). The nonlinear robust term is expressed as $v_{r2} = -Z(q, \dot{q}, t) = -(1/4\varpi)\rho^2s = -k_{r2}s$.

Theorem 1: It is assumed that the adaptive function is designed as

$$\tau = \psi_r s. \quad (18)$$

Therefore, the following theorem can be obtained by using the ARC law (13).

P1: Generally speaking, the control inputs and other signals of the system are bounded. In addition, the positive definite energy function $V(t)$ can be inferred to satisfy the following boundary constraints:

$$V(t) \leq \exp(-\zeta t)V(0) + \varpi[1 - \exp(-\zeta t)]/\zeta \quad (19)$$

where $\zeta = 2\chi_{\min}(k_{r1})/\delta_2$, and $\chi_{\min}(\bullet)$ represents the minimum eigenvalues of the matrix $\bullet$. Obviously, $\chi_{\min}(k_{r1})$ is a positive real number.

P2: It is assumed that only parameter uncertainties exist in the system after a finite time t_0 (i.e., $\tilde{\Delta} = 0 \quad \forall t \geq t_0$). Then, the final error convergence to zero can also be theoretically guaranteed except for conclusion P1. That is, when $t \rightarrow \infty$, $e \rightarrow 0$, and $s \rightarrow 0$.

Proof: It is known that the parameter is bounded from (8) and (11), we have

$$\frac{\delta_1 \|s\|^2}{2} \leq V(t) \leq \frac{\delta_2 \|s\|^2}{2}. \quad (20)$$

Substituting (13) and (16) into (14), we have

$$\begin{aligned} \dot{V}(t) &= -s^T k_{r1}s + s^T \left\{ v_{r2} - [\psi_r(q, \dot{q}, t)\tilde{\theta}_r - \tilde{\Delta}] \right\} \\ &\leq -\chi_{\min}(k_{r1})\|s\|^2 + \varpi \\ &= -\zeta V + \varpi. \end{aligned} \quad (21)$$

Conclusion P1 is clearly proved and the boundedness of s can be inferred from (11). The sliding surface function (10) can be expressed by the Laplace transform as $G_s(s) = 1/(s + \kappa)$. Obviously, it is a stable convergent transfer function, and the convergence rate is κ . Namely, when s is small or tends to zero, σ is also small or tends to zero. Therefore, the stable transfer function $G_s(s)$ shows that if s is bounded, it guarantees that σ and $\dot{\sigma}$ is also bounded. Furthermore, the desired trajectory q_d is

obviously bounded and differentiable, q is also bounded with the boundedness of q_d and σ . The property of the adaptive law (24) guarantees the boundedness of $\hat{\theta}$, and the control input of v is also bounded. Finally, it follows that all signals are bounded.

Considering the conclusion of P2 (i.e., $\tilde{\Delta} = 0$) $\forall t \geq t_0$, a new energy function $V_2(t)$ is defined as

$$V_2(t) = V(t) + \frac{1}{2} \tilde{\theta}_r^T \Gamma_r^{-1} \tilde{\theta}_r. \quad (22)$$

Substituting (24) and (27) into (21), the following formula can be derived:

$$\begin{aligned}\dot{V}_2(t) &\leq -s^T k_{r1}s - s^T \psi_r(q, \dot{q}, t) \tilde{\theta}_r + \tilde{\theta}_r^T \Gamma_r^{-1} \dot{\hat{\theta}}_r \\ &= -s^T k_{r1}s + \tilde{\theta}_r^T [\Gamma_r^{-1} \dot{\hat{\theta}}_r - \psi_r(q, \dot{q}, t)s] \\ &\leq -s^T k_{r1}s\end{aligned}\quad (23)$$

where $s \in \mathcal{L}_2 \cap \mathcal{L}_\infty$, and it can be easily concluded that s is uniformly continuous. The conclusion of P2 can be obviously obtained from Barbalat's lemma, that is, when $t \rightarrow \infty$, $e \rightarrow 0$, and $s \rightarrow 0$. ■

B. Discontinuous Projection

It can be seen from the controller design (13) that the estimation error $\hat{\theta}_r$ can be reduced as much as possible and the stability of the system can be improved by selecting an appropriate adaptive law. In order to ensure the boundedness of parameter estimation and avoid the conservatism of traditional adaptive law, an adaptation law with discontinuous projection in [29] is designed as

$$\dot{\hat{\theta}}_r = \text{Proj}_{\hat{\theta}_r}(\Gamma_r \tau) \quad (24)$$

where Γ_r is a diagonal matrix of positive constant and represents adaptive learning rate. τ denotes the adaptation function, which is determined by (18). The projection mapping $\text{Proj}_{\hat{\theta}_r}(\ast) = [\text{Proj}_{\hat{\theta}_{r1}}(\ast_1), \dots, \text{Proj}_{\hat{\theta}_{r(8+4m)}}(\ast_{(8+4m)})]^T$ is written as

$$\text{Proj}_{\hat{\theta}_{ri}}(*_i) = \begin{cases} 0, & \text{if } \hat{\theta}_{ri} \geq \theta_{ri \max} \text{ and } *_i > 0 \\ 0, & \text{if } \hat{\theta}_{ri} \leq \theta_{ri \min} \text{ and } *_i < 0 \\ *_i, & \text{else.} \end{cases} \quad (25)$$

Invoking [30], the following two properties can be easily obtained:

$$i. \quad \hat{\theta}_r(t) = \bar{\Omega}_{\theta r} \triangleq \left\{ \hat{\theta}_r : \theta_{r \min} \leq \hat{\theta}_r \leq \theta_{r \max} \right\} \quad (26)$$

$$ii. \quad \tilde{\theta}_r(t) \left(\Gamma_r^{-1} \text{Proj}_{\hat{\theta}_r} (\Gamma_r \tau) - \tau \right) \leq 0 \quad \forall \tau \quad (27)$$

where the parameter estimation $\hat{\theta}_r$ always falls in the closed region $\bar{\Omega}_{\theta r}$.

IV. EXPERIMENTAL VERIFICATION

A. Experiment Setup

The comparative experiments have been successfully completed on the gantry stage manufactured by the Akribis company, as shown in Fig. 1. Two redundant linear motors jointly drive the beam in the Y direction and one linear motor drives the moving table in the X direction. Three encoders

with an accuracy of 50 nm are installed along the linear guide rail to measure the absolute position information of the three servo drive shafts. The velocity signal can be obtained by differential calculation of the measured position information. The whole closed-loop system is controlled by industrial computer in real time, and EtherCAT communication protocol is adopted to ensure information transmission. The sampling interval of the whole program is $T = 1$ ms.

B. Parameter Selections

According to the reference manual, the mass of the beam is 25 kg and the distance between the magnetic poles can be measured as 43 mm. Square wave excitation signal is used as the input of the system, and then the position, speed and acceleration signals of the output are collected by the terminal computer. After the second-order filtering function $H_f(s) = 1/(\Xi_f + 1)^2$ processing, the least square method was used to get the nominal values of the parameters of the dynamics model. In addition, considering that the results obtained may be affected by the existence of unknown disturbances in the actual environment, the estimated values of model parameters and their upper and lower bounds are, respectively, set as

$$\begin{aligned}\hat{\theta}_r(0) &= [0.12, 0.12, 0.03, 0.03, 0.03, 0.03, 0, 0]^T \\ \theta_{r\min} &= [0.1, 0.1, 0.02, 0.02, 0.01, 0.01, -0.01, -0.01]^T \\ \theta_{r\max} &= [0.2, 0.2, 0.04, 0.04, 0.05, 0.05, 0.01, 0.01]^T\end{aligned}$$

where the harmonic components of y-axis cogging force are mainly distributed on $m = 1, 6$, and 12 in [26]. The initial and boundary values are set to 0 and ± 0.2 , respectively, by offline parameter identification in $\hat{\theta}_r$.

Several comparative experiments are presented below to test the effectiveness of the proposed control strategy. It should also be noted that when the controller parameters have the same meaning, the controller parameters should be kept consistent to improve the reliability of the experiment.

CI: Traditional master-slave structure control strategy based on ARC. The robust control term in (15) is chosen as a sufficiently large feedback gain K_r , which synthesizes k_{r1} and k_{r2} to simplify the design of the controller. The controller gains are designed as $K_r = \text{diag}[1.5, 1.5]$ and $\kappa = \text{diag}[22, 22]$. The nonlinear function $S_f(\dot{q})$ is selected as $(2/\pi) \arctan(500\dot{q})$. Letting $\Gamma_r = \text{diag}[10, 10, 20, 20, 10, 10, 100, 100]$, which represents the adaptive learning rate of parameters.

C2: Improved master slave structure control strategy based on ARC. Select the same parameters as C1.

C3: ARC control strategy with cogging force compensation based on C2. Select the same parameters as C1, and let $\Gamma_r = \text{diag}[10, 10, 20, 20, 10, 10, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 100, 100]$ and $\hat{\theta}_r(0) = [0.12, 0.12, 0.03, 0.03, 0.03, 0.03, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]^T$.

In addition, three different settings will be designed to verify the robustness of the proposed control strategy.

Set1: The moving stage is statically placed in the center of the beam.

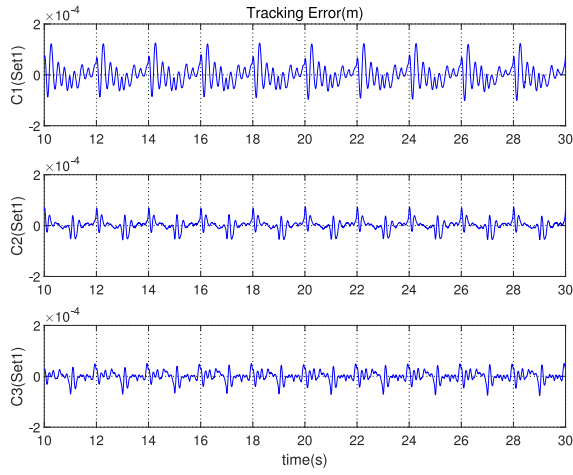


Fig. 3. Tracking errors of three control strategies in Set1.

Set2: The moving stage is located at one end of the slave servo unit of the beam.

Set3: The moving stage reciprocates along the direction of beam, and its movement instruction is the same as that of the beam.

C. Experimental Results

In the experiment, the desired trajectory of beam is chosen as $y_d = 10^{-2}(1 - \cos(\pi t))$. In addition, in order to avoid the introduction of large error in the initial stage of the experiment, the performance index 10-s later was selected to evaluate the experimental results.

Performance Evaluation Index:

$\|e\|_2 = \sqrt{(\int_{10}^T |e(t)|^2 dt)/(T - 10)}$ stands for the root mean square of error, which reflects the average effect of motion control performance.

$\|e\|_\infty = \max(|e(t)|)$ denotes the maximum absolute value of error, which reflects the worst transient performance of motion control system.

$\|u\|_{rms} = \sqrt{(\int_{10}^T (|u_1(t)| + |u_2(t)|)^2 dt)/(T - 10)}$ represents the root mean square value of the absolute value of the control input, which is used to display the strength of control input required for normal movement of the system.

Case I: The experimental results of different control strategies in Set1 are indicated in Fig. 3. It can be seen from the figure that the tracking performance of the improved master-slave structure has been greatly improved. Compared with C1, the amplitude fluctuation of the latter two methods is more stable. It is clear to know from Table I, the value of $\|e\|_2$ for C2 is decreased from 41.424 to 21.261 μm and the value of $\|e\|_\infty$ is reduced from 125.170 to 74.195 μm . This is because the proposed master-slave control adopts a new structure, which makes the driving unit follow the main driving unit faster and avoids the hysteresis effect. The connection between the master and slave servo units is established through error dynamic coupling, which makes the system respond to the uncertain disturbance in the environment and adjust in time.

Fig. 4 shows the experimental results of the synchronization error under different control strategies. Similar to the tracking

TABLE I
PERFORMANCE INDEXES IN SET1

Control Strategies	$\ e_y\ _2$	$\ e_y\ _\infty$ μm	$\ e_{sync}\ _2$	$\ u\ _{rms}$ V
C1(Set1)	41.424	125.170	3.3280	0.0412
C2(Set1)	21.261	74.195	2.3584	0.0431
C3(Set1)	20.851	75.982	2.0999	0.0428

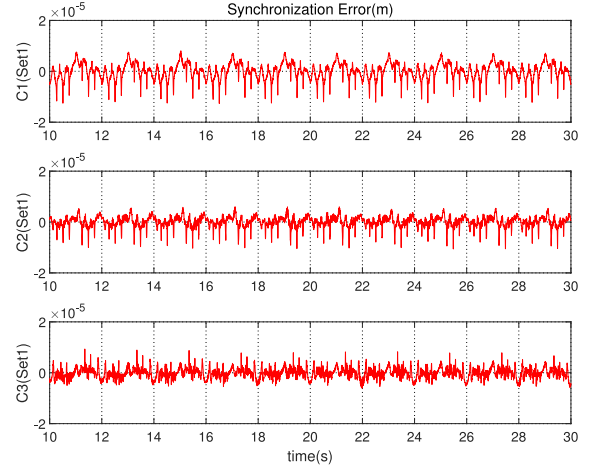


Fig. 4. Synchronization errors of three control strategies in Set1.

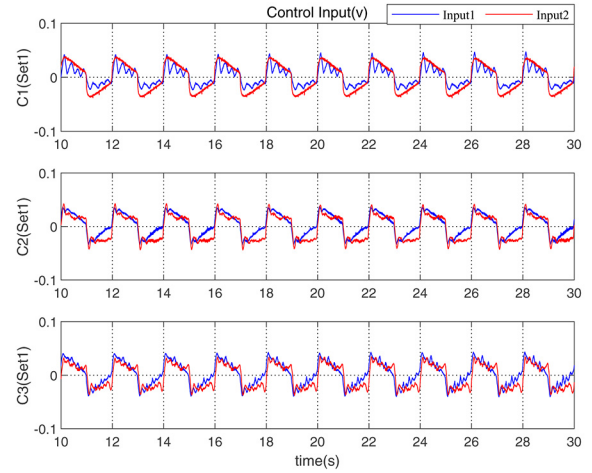


Fig. 5. Control inputs for the three control strategies in Set1.

error results, the synchronization performance of the improved master-slave structure is greatly improved. In addition, the amplitude of the synchronization error of C3 is smaller than that of C2 when the cogging force compensation is used. The value of $\|e_{sync}\|_2$ is decreased from 2.3584 to 2.0999 μm in Table I, which directly illustrates the effectiveness of cogging force compensation in synchronous control.

The results of control inputs under three different control strategies are illustrated in Fig. 5. The calculated results show that the linear motor control inputs are approximately equal to each other. However, considering the synchronization performance between the two axes, the control inputs of C2 and C3 achieve a high level of consistency in shape, which also reflects the advantage of using the proposed control algorithm.

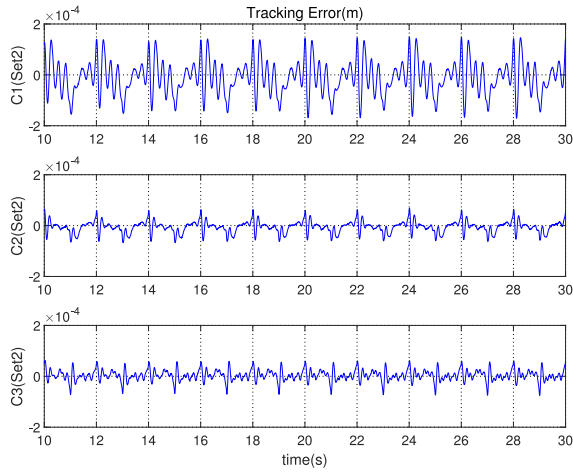


Fig. 6. Tracking errors of three control algorithms in Set2.

TABLE II
PERFORMANCE INDEXES IN SET2

Control Strategies	$\ e_y\ _2$	$\ e_y\ _\infty$ μm	$\ e_{sync}\ _2$	$\ u\ _{rms}$ V
C1(Set2)	67.557	171.050	4.0566	0.0434
C2(Set2)	23.528	68.716	2.2172	0.0446
C3(Set2)	22.223	75.482	2.0893	0.0440

Case II: Fig. 6 shows the tracking error results of different strategies when the moving stage is on the side of the slave servo unit. It can be seen from the plots of C1 that the load of the moving table deteriorates the hysteresis effect of the slave servo unit, which leads to further expansion of the tracking error. Moreover, the results in Table II show that the value of $\|e_y\|_2$ has increased to 67.557 μm and the value of $\|e_y\|_\infty$ has risen to 171.050 μm for C1, which leads to system instability and seriously affects the control accuracy of the system. In contrast, the values of $\|e_y\|_2$ and $\|e_y\|_\infty$ for C2 and C3 have little changed under the same experimental conditions. In other words, the proposed improved master-slave scheme can minimize the impact of the load changes and improve the tracking performance of the system. Although the tracking error trend of C2 and C3 is almost consistent with Set1, it should be pointed out that the amplitude of C3 is always smaller than C2. The results show that the control scheme of C3 still has good robustness under load disturbance.

Fig. 7 shows the synchronization error of the gantry stage in Set2. The curves in the figure are very similar to Set1 and the calculated results are detailed in Table II. It can be seen from the calculation results of the performance index that the value of $\|e_{sync}\|_2$ for C1 has increased from 3.3280 to 4.0566 μm and the values of $\|e_{sync}\|_2$ for C2 and C3 are almost the same as Set1, which further indicates that the improved master-slave structure has good synchronization control performance. In particular, the synchronization error of C3 is still less than that of C2 even under load interference, which illustrates that non-linear cogging force compensation plays an important role in improving the synchronization accuracy of the gantry system.

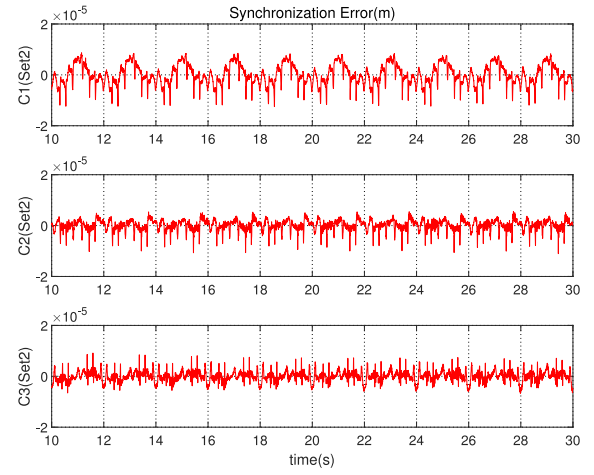


Fig. 7. Synchronization errors of three control strategies in Set2.

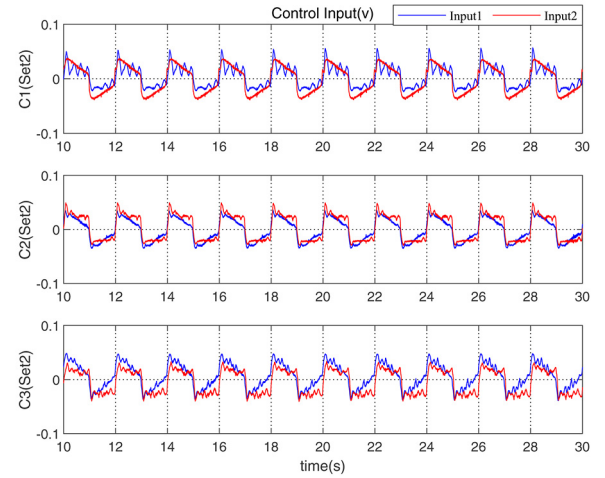


Fig. 8. Control inputs for the three control strategies in Set2.

Fig. 8 displays the control inputs of two linear motor drivers. The values of $\|u\|_{rms}$ for C1, C2, and C3 are relatively close, but it can be seen that the inputs of the two controllers fluctuate greatly for C1. This is because the response lag of the servo axis is more serious when the centroid of the beam is close to the slave servo unit, and the performance of the system is more susceptible to interference. In this case, the master servo drive shaft always drags the slave servo unit, which severely affects the system performance and ultimately causes mechanical damage to the platform. The control inputs for C2 and C3 are almost harmonized, which indicates the stronger robustness of the system.

Case III: Fig. 9 indicates the tracking error generated when the moving stage reciprocates along the beam direction. Obviously, the amplitude of tracking error of the first group experiments in the figure is still the largest compared with the other groups. Compared with Set2, the amplitude of curves of C2 and C3 in the comparative experiments is still in a small range, and the results in Table III show that the value of $\|e_y\|_2$ for C1 is reduced from 67.557 to 48.936 μm . The results show that static load has a great influence on slave servo system, and the ARC algorithm makes the system better

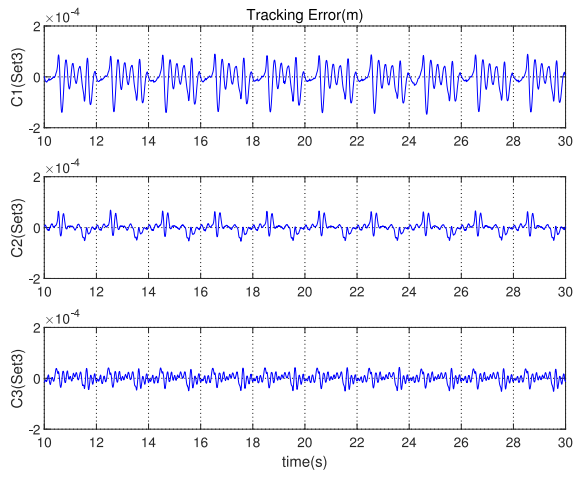


Fig. 9. Tracking errors of three control strategies in Set3.

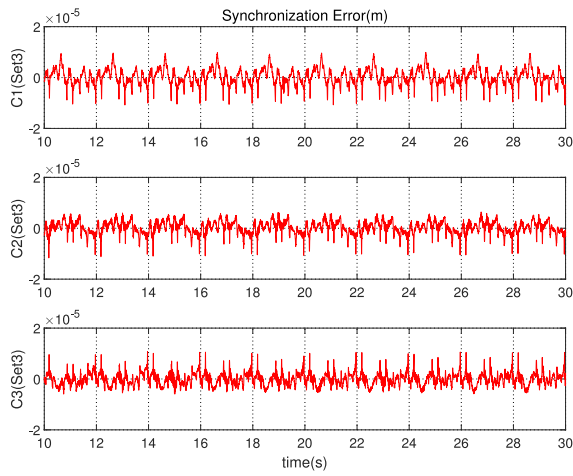


Fig. 10. Synchronization errors of three control strategies in Set3.

cope with the influence of parameter uncertainties caused by environmental disturbance. In addition, the values of $\|e_y\|_2$ and $\|e_y\|_\infty$ for C2 and C3 change little compared with the former sets of experiments. The experimental results illustrate that the excellent tracking performance of the system can be effectively achieved by adopting the proposed control strategy, which is consistent with the predicted results.

The synchronization error results of the three control strategies under the influence of moving stage movement along the beam are shown in Fig. 10. In the figure, the synchronization error curve of C3 has small fluctuation and is relatively smooth compared with C1 and C2. The influence of nonlinear factors in the motion process is relatively suppressed and the precision of beam is improved by compensating the cogging force. Similarly, the values of $\|e_{sync}\|_2$ for C2 and C3 are slightly different from those of the former two groups in Table III, but they are still less than the experimental results of C1, and the performance of C3 remains optimal. In conclusion, the proposed control strategy makes the dual-drive gantry system deal with the influence of external factors and parameter changes, and has good synchronization control performance.

TABLE III
PERFORMANCE INDEXES IN SET3

Control Strategies	$\ e_y\ _2$	$\ e_y\ _\infty$ μm	$\ e_{sync}\ _2$	$\ u\ _{rms}$ V
C1(Set3)	48.936	147.260	3.2603	0.0469
C2(Set3)	20.192	68.553	2.6456	0.0432
C3(Set3)	18.308	51.542	2.5582	0.0443

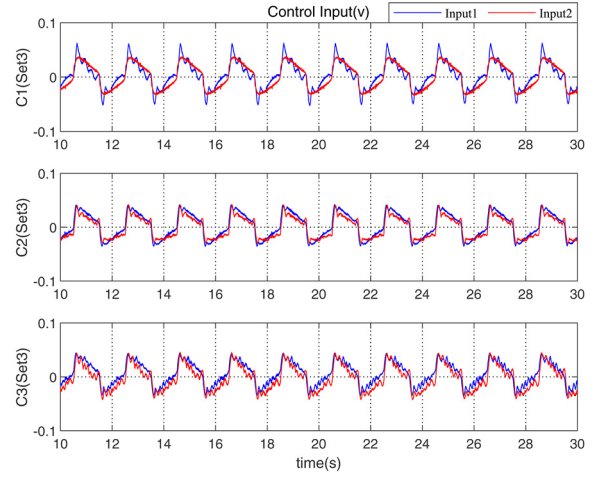


Fig. 11. Control inputs for the three control strategies in Set3.

Fig. 11 indicates the experimental results of control input under three control strategies. As we can see, the two control inputs of C1 are offset from each other and have spikes in shape. In contrast to C1, the two control inputs of C2 and C3 are roughly identical and basically symmetrical. In other words, when the gantry platform moves along the desired trajectory and is disturbed, the improved master-slave synchronous control strategy can make the system timely adjust and keep the system as stable as possible. Obviously, the result of control input in Table III also confirms this inference. The advantages of the improved master-slave structure compared with the traditional master-slave structure are verified, and the synchronization performance of the system will be better after adding the compensation of the cogging force compensation.

In order to make the comparison more apparent, the performance indexes of all experimental results are visualized and displayed in Fig. 12. The results in the figure clearly show that the performance index of control strategy C2 is better than that of C1. It shows that the improved master-slave synchronization scheme can keep the stable operation of the system and obtain excellent tracking and synchronization performance even if there are static or dynamic disturbances in the gantry platform system. In addition, the control strategy C3 has stronger robustness and better synchronization control accuracy than C2 even under the influence of parameter uncertainties and external disturbances, which fully verifies the practicability and effectiveness of the improved master-slave synchronization control scheme with cogging force compensation. It is worth mentioning that compared with the more advanced control strategy in [11], the synchronization control

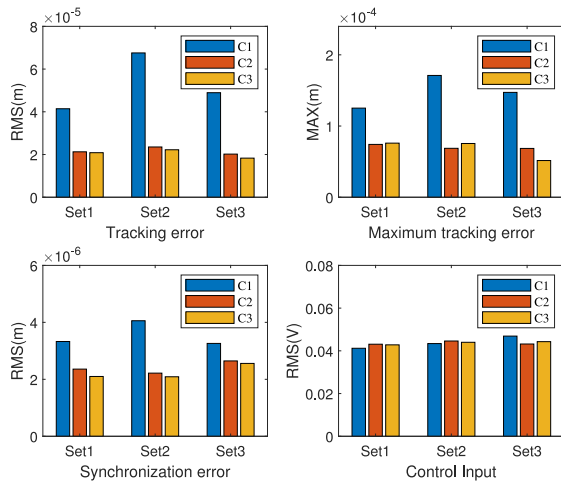


Fig. 12. Block diagram of performance index comparison.

method in this article achieves similar tracking and synchronization performance, but is significantly superior to the former in terms of practical operation and computational complexity. In industrial applications, the simplicity and easy implementation of the control algorithm is a very important concern. It can effectively reduce the failure rate of equipment and improve the reliability of the system, which is also the advantage of this method in practical application.

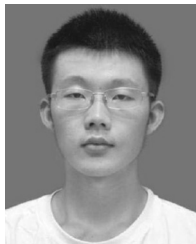
V. CONCLUSION

It is very important to carefully consider the complexity of controller design and the disturbance of nonlinear factors in the system to achieve good control accuracy of the dual-drive gantry stage. In this article, considering the above two key factors, an improved master-slave synchronization control strategy is proposed for the dual-drive gantry stage to obtain excellent tracking and synchronization performance. The proposed synchronization control scheme is not only simple in structure and easy to implement, but also can obtain good synchronization accuracy when applied to the industrial gantry platform system. The error-coupled dynamic model is established to minimize the hysteresis problem of traditional master-slave control. The adaptive algorithm is designed to deal with the parameter uncertainties in the system and compensate the model, while the robust control deals with the influence of unmodeled dynamics and external disturbances. In addition, the nonlinear cogging force model is established and compensated for the dual-drive gantry system to reduce the influence of nonlinear factors and further improve the accuracy of synchronization. Finally, the system achieves good tracking and synchronization performance under different conditions. By comparing the experimental results, the superiority of the proposed control strategy has been verified.

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